Can Modified SCCS Detect Increased SIDS Risk Following Vaccination? A Simulation Study

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Previously, I explained why the retraction/removal of the Miller SIDS paper—which claimed that sudden infant death syndrome (SIDS) increases after vaccination—was justified. The paper's central analysis relied on incorrect statistical assumptions about VAERS data, rendering its analysis, results, and conclusions invalid.

(https://x.com/jsm2334/status/2057919431519694866)

In a follow-up post, I discussed why VAERS is not the ideal data source for answering this question. Because VAERS is a passive surveillance system with reporting bias and no control group, it cannot directly determine whether vaccination increases SIDS risk. Instead, the best evidence comes from population-based data sets that include every SIDS death, vaccination dates, dates of death, and the interval between vaccination and death.

(https://x.com/jsm2334/status/2067374044845089174)

I highlighted six studies from five countries spanning three continents that addressed this question using a modified self-controlled case series (SCCS) design. SCCS is particularly well suited for questions such as:

Is the risk of SIDS increased immediately following vaccination?

Its key advantage is that each infant serves as his or her own control, eliminating confounding from factors that differ systematically between vaccinated and unvaccinated infants, including the healthy vaccinee effect. Because SIDS is a terminal event, these studies used the modified SCCS method developed by Farrington, which extends the original SCCS framework to appropriately handle event-dependent exposures and deaths.

https://pubmed.ncbi.nlm.nih.gov/18499654/

Taken together, these six studies found no evidence that vaccination increases the risk of SIDS, either overall or immediately following vaccination.

Following those posts, several people questioned whether the modified SCCS design is actually capable of detecting vaccine-associated SIDS if such an effect truly existed. One individual even speculated—without evidence—that Farrington had been "commissioned" to develop a method that "hides vaccine injury"

https://x.com/Jikkyleaks/status/2069599158001115251

and others similarly questioned whether the modified SCCS approach was appropriate.

https://x.com/jsm2334/status/2070259110050963943?s=20

Rather than debating the issue theoretically, I decided to test it directly.

I conducted a simulation study designed to answer two straightforward questions:

1. Does the modified SCCS incorrectly conclude vaccine harm when no increased risk exists?
2. If vaccination truly increases the short-term risk of SIDS, can the modified SCCS detect it?

This article presents that simulation study, along with the R code used to generate the simulated data, fit the modified SCCS models, and summarize the results.

Executive summary

If you don't want to read the technical details, here's the bottom line.

Using realistic age distributions for both SIDS incidence and vaccination timing, a population of five million infants, and an overall SIDS incidence of approximately 1 in 5,000, I evaluated the modified SCCS both with and without adjustment for age.

The results were clear.

- When age was appropriately adjusted for, the modified SCCS rarely produced false-positive findings. In the null simulations (no vaccine effect), vaccine harm was incorrectly concluded in only about 7% of simulations, close to the nominal 5% expected using a $p<0.05$ significance threshold.
- When a true increase in post-vaccination risk was introduced, the method had high power to detect it. For example, it had approximately 81% power to detect a 50% increase in SIDS risk during the three days following vaccination.
- In contrast, failing to adjust for age produced completely invalid results. Because both SIDS incidence and vaccination timing are strongly age-dependent, the unadjusted analysis falsely concluded vaccine harm essentially 100% of the time, even when no vaccine effect existed.
- I also evaluated all age-group definitions used in the six published SCCS studies. Although age groups that were excessively wide produced some upward bias in the estimated vaccine effect, the results were generally robust provided the age categories were reasonably chosen.

The principal conclusion is straightforward:

The modified SCCS performs well for this question, provided age is appropriately controlled.

Simulation design

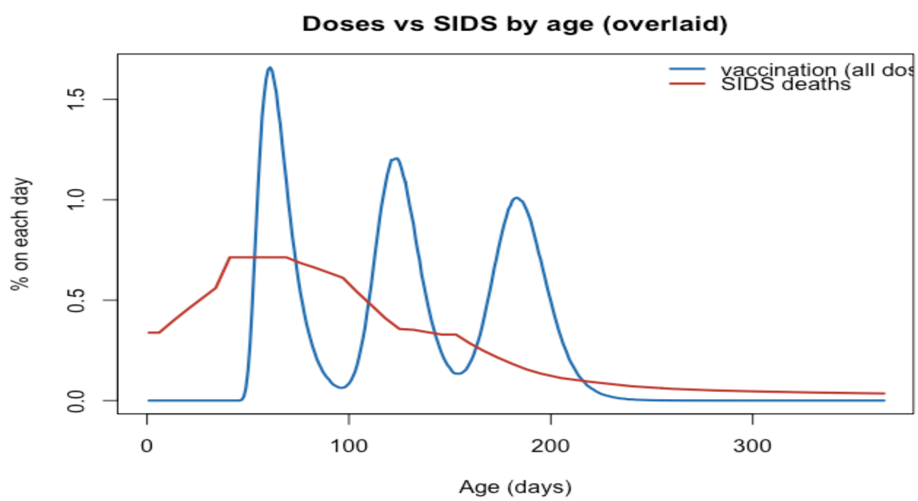
To evaluate the method, I generated a virtual population of infants, simulated the timing of doses 1, 2, and 3 of DTaP (approximately 2, 4, and 6 months of age), and then simulated which infants experienced SIDS and when.

The simulation made the following assumptions:

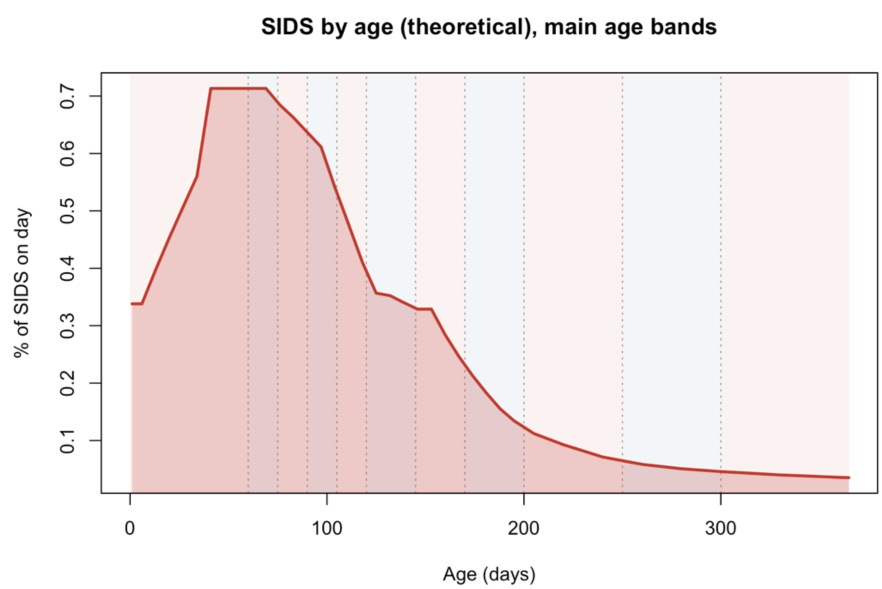
1. A population of five million infants followed throughout the first year of life.
2. An overall SIDS incidence of approximately 1 in 5,000, consistent with the published literature.
3. The baseline age distribution of SIDS (assuming no vaccine effect) was estimated by fitting the observed age-incidence curve from one published study whose overall shape has been replicated in other population-based studies.
4. Vaccination timing for dose 1 was based on publicly available UK vaccination data. Doses 2 and 3 occurred an average of 60 days later, with realistic variability matching observed vaccination schedules.
5. Vaccine effects were modeled as increased SIDS risk during the three days following each vaccine dose. Separate simulations assumed incidence rate ratios (IRRs) of 1.0 (no effect), 1.25, 1.5, 1.75, and 2.0.
6. Each scenario was simulated 100 times. For each setting I summarized:

- Mean estimated IRR (bias)
- Proportion of simulations with p<0.05 (false-positive rate when IRR=1; statistical power otherwise)
- Coverage of the 95% confidence interval (which should be close to 95% for a well-calibrated method)

The figure below shows the simulated background age distribution of SIDS together with the distribution of vaccination ages.



In the primary analysis, I fit the modified SCCS using the three days following each vaccine dose as the risk period. Follow-up began on day 1 and continued through day 365. Pre-vaccination SIDS cases were retained because they help estimate the underlying age-specific baseline incidence. Age adjustment used categories chosen so that baseline SIDS incidence was approximately constant within each age band, as indicated by the alternating blue and red bands in the following figure:



Results

The simulation results are shown below.

Primary Simulation

IRR (true)	Mean(IRR)	Prob(p<0.05) Power (Type 1 error*)	Mean Coverage of 95% CI
1.0	1.06	0.07*	93%
1.25	1.24	0.37	93%
1.50	1.45	0.81	98%
1.75	1.65	0.98	97%
2.00	1.85	1.00	91%

For the null scenario (IRR = 1.0):

- Only about 7% of simulations incorrectly concluded vaccine harm, close to the expected 5% false-positive rate.
- The average estimated IRR was close to 1.0, indicating minimal bias.
- Ninety-three percent of 95% confidence intervals contained the true value, close to the nominal 95% target.

When vaccination truly increased SIDS risk:

- The modified SCCS detected the effect with high statistical power.
- Power was approximately:
  - 37% for a 25% increase (IRR = 1.25)
  - 81% for a 50% increase (IRR = 1.50)
  - 98% for a 75% increase (IRR = 1.75)
  - 100% for a doubling of risk (IRR = 2.0)
- Estimated IRRs remained close to their true values, indicating little bias.
- Confidence interval coverage remained close to the nominal 95%, indicating good statistical calibration.

Overall, these simulations show that the modified SCCS is well suited for evaluating whether SIDS risk increases immediately after vaccination. When age is modeled appropriately, the method has a low false-positive rate, good calibration, and substantial power to detect clinically meaningful increases in risk if they truly exist.

What happens if we don't adjust for age?

To illustrate how essential it is to account for age, I repeated the simulation using the identical data-generating process but omitted age from the modified SCCS model. In other words, the model was not allowed to adjust for the strong underlying relationship between age and baseline SIDS risk. All other aspects of the simulation remained unchanged.

The results are shown below.

No Adjustment for Age Confounding

IRR (true)	Mean(IRR)	Prob(p<0.05) Power (Type 1 error*)	Mean Coverage of 95% CI
1.0	2.65	1.00*	0%
1.25	3.08	1.00	0%
1.50	3.59	1.00	0%
1.75	4.58	1.00	0%
2.00	8.44	1.00	0%

The consequences were dramatic:

- **The model concluded statistically significant vaccine harm in 100% of simulations**, including the null scenario (IRR = 1.0) in which vaccination had **no effect whatsoever** on SIDS risk.
- **Estimated incidence rate ratios (IRRs) were severely inflated**, substantially overstating the magnitude of vaccine-associated risk in every scenario.
- **Coverage of the 95% confidence intervals was 0%**, meaning the intervals never contained the true effect. This indicates the model was completely statistically miscalibrated.

These failures occur because both **SIDS incidence** and **vaccination timing** are strongly age-dependent. SIDS risk peaks during the first few months of life and then declines rapidly, while routine infant vaccinations are also administered at specific ages. Without adjusting for age, the SCCS model incorrectly attributes this natural overlap to vaccination itself, creating the appearance of increased risk even when none exists.

This simulation demonstrates that **adjusting for age is not an optional refinement—it is essential for valid inference**. When age is modeled appropriately, the modified SCCS performs well, exhibiting low bias, good statistical calibration, and high power to detect genuine increases in risk. Without age adjustment, however, the analysis is fundamentally invalid, producing false evidence of vaccine harm even when no such harm exists.

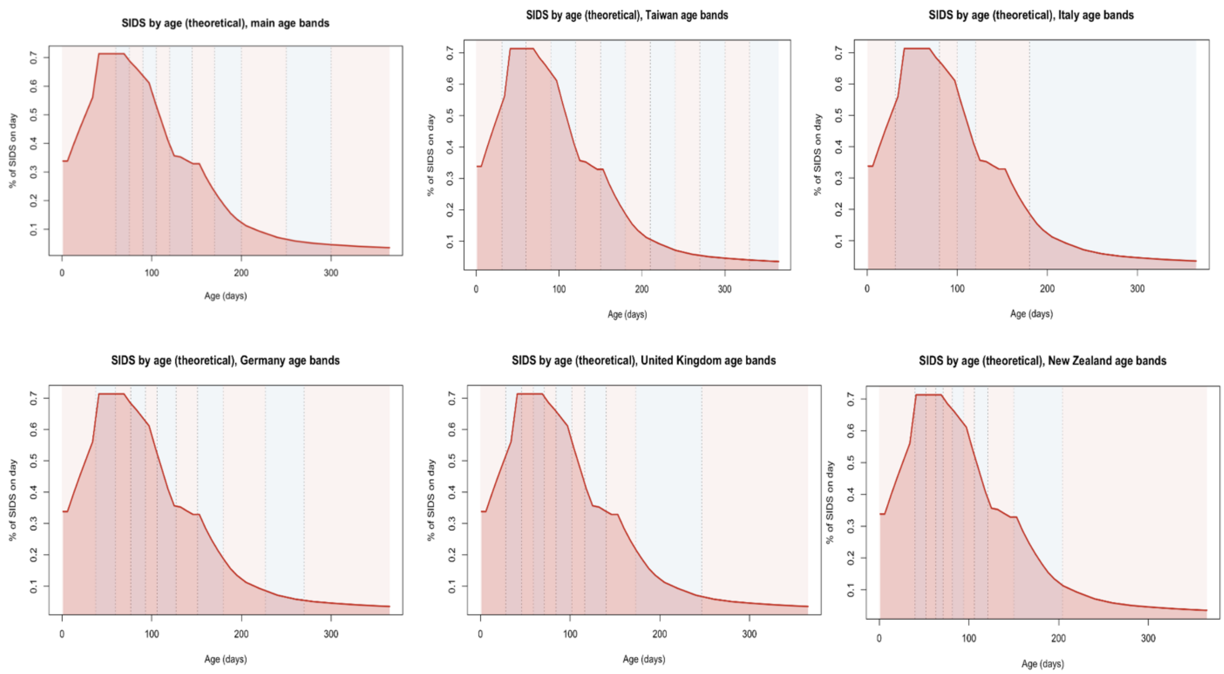
Assessing Robustness to the Choice of Age Bins

The six published modified SCCS studies of SIDS following vaccination all adjusted for age, but they did not use identical age categories. This raises an important question:

How sensitive are the results to the particular choice of age bins?

Ideally, age categories should be chosen so that the baseline incidence of SIDS is approximately constant within each age bin. However, there is no unique way to accomplish this, so it is important to determine whether reasonable alternative choices materially affect the results.

To investigate this, I repeated the primary null simulation (IRR = 1.0) using the age-group definitions from each of the six published studies. The figures below illustrate the age bins used in the studies from Taiwan, Italy, Germany, the UK, and New Zealand, shown as vertical lines over the underlying SIDS age-incidence curve.



The table below summarizes the simulation results for each set of age bins.

Null Simulation, age bins by study

IRR (true)	Mean(IRR)	Prob(p<0.05) Type 1 error*	Mean Coverage of 95% CI
Taiwan	1.06	0.07*	93%
Italy	1.11	0.10*	90%
Germany	1.03	0.06*	94%
UK	1.07	0.06*	94%
New Zealand	1.04	0.06*	94%

Results

Overall, the modified SCCS performed consistently across all of the published age categorizations.

- The false-positive rate ranged from **6% to 10%**, close to the nominal 5% level expected under the null hypothesis.
- Mean estimated IRRs ranged from **1.03 to 1.11**, indicating only modest upward bias.
- Coverage of the 95% confidence intervals ranged from **90% to 94%**, reasonably close to the nominal 95% level.

These findings indicate that the modified SCCS is **robust to reasonable choices of age bins**, provided age is modeled with sufficient resolution to capture the rapidly changing baseline risk of SIDS during infancy.

Among the published studies, the Italian analysis used the widest age categories. Because some of these bins span ages over which baseline SIDS incidence changes substantially, that specification produced the largest upward bias (mean IRR = 1.11) and the highest false-positive rate (10%). Although these values remain reasonably close to the nominal targets, they illustrate an important practical point: **overly wide age bins can leave residual age confounding, leading to modest inflation of estimated vaccine effects and false-positive rates.**

Taken together, these simulations suggest that the conclusions of the published SCCS studies are **not highly sensitive to the specific age-group definitions that were used**, while also reinforcing the importance of choosing age categories that adequately capture the strong age dependence of SIDS incidence.

Conclusion

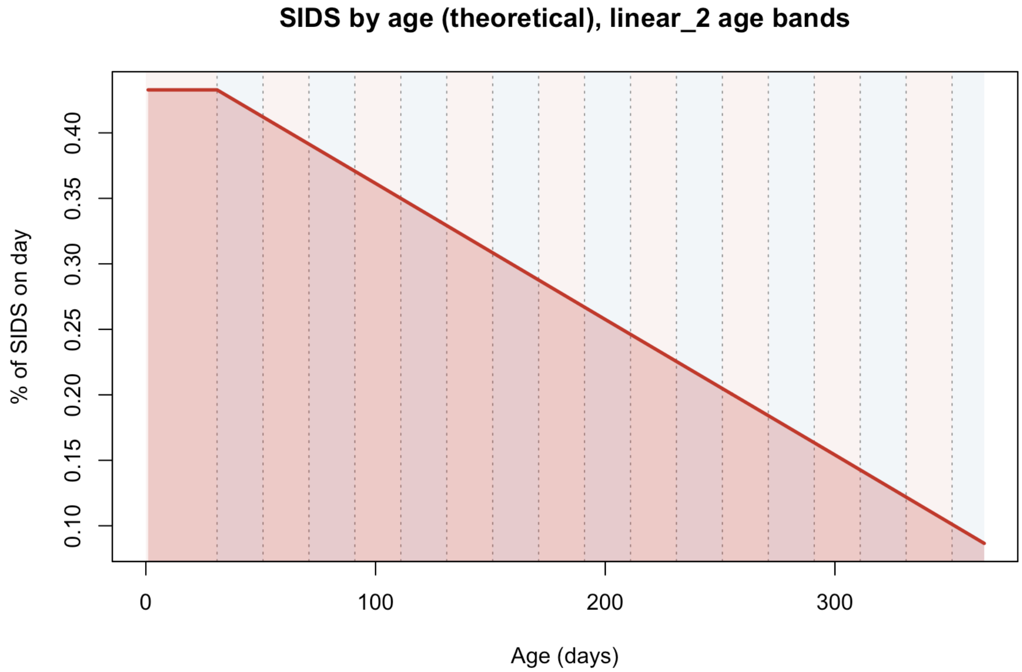
The modified self-controlled case series (SCCS) has been questioned by some as an inappropriate method for investigating whether SIDS risk increases immediately following vaccination. These simulations provide little support for that criticism.

Using realistic patterns of SIDS incidence by age and vaccination timing, the modified SCCS performed well when age was appropriately modeled. It exhibited low bias, good statistical calibration, and substantial power to detect clinically meaningful increases in post-vaccination SIDS risk. In contrast, omitting age adjustment produced completely invalid results, generating strong but entirely spurious evidence of vaccine harm because of the well-known confounding created by the strong age dependence of both SIDS incidence and routine infant vaccination.

Assessing Sensitivity to the Assumed Age Distribution of SIDS

It has been suggested that the age-specific SIDS incidence curve observed in the Traversa study may not accurately reflect the true background incidence of SIDS in an unvaccinated population, and that the strong performance of the modified SCCS in these simulations might depend on this particular choice. As an alternative, it was proposed that baseline SIDS incidence instead declines approximately linearly with age.

To evaluate this, I ran an additional sensitivity analysis that replaced the Traversa age distribution with a linearly declining age-specific SIDS incidence. As before, I chose age bins narrow enough that baseline SIDS incidence was approximately constant within each category. The resulting age distribution and bins are shown below.



Using this alternative distribution, I repeated the simulation settings described previously, fitting the modified SCCS both with and without age adjustment. The results for the modified SCCS with age adjustment are shown below.

Linear Age Effect

IRR (true)	Mean(IRR)	Prob(p<0.05) Power (Type 1 error*)	Mean Coverage of 95% CI
1.0	0.99	0.05*	95%
1.25	1.21	0.24	94%
1.50	1.39	0.61	92%
1.75	1.59	0.89	90%
2.00	1.77	0.95	86%

The findings closely mirror those from the original analysis:

- Under the null, the IRR was nearly unbiased, with a nominal false-positive rate and nominal coverage.

- When vaccines truly increased SIDS risk, the modified SCCS reliably detected harm: power was strong for 100% and 75% increases (IRR = 2.00 and 1.75, with 95% and 89% power, respectively) and reasonable for a 50% increase (IRR = 1.50, 61% power).
- IRRs remained close to their true values, with modest negative bias.
- Confidence interval coverage (86%–94%) stayed reasonably close to nominal.

The results without any age adjustment are shown below.

Linear Age, No Adjustment for Age Effect

IRR (true)	Mean(IRR)	Prob(p<0.05) Power (Type 1 error*)	Mean Coverage of 95% CI
1.0	1.33	0.43*	57%
1.25	1.63	0.85	58%
1.50	1.88	0.99	63%
1.75	2.15	1.00	67%
2.00	2.39	1.00	70%

As before, omitting the age effect has serious consequences: the false-positive rate rose to 43%, meaning the model would frequently conclude vaccine harm when none existed, and the depressed coverage indicates poor statistical calibration.

Overall, these simulations show that the modified SCCS continues to perform well under a linearly declining background SIDS risk, provided age adjustment is used. Its good performance does not depend on the specific Traversa-motivated background distribution adopted in the original simulations.

Simulation R Scripts to Reproduce Results

The R scripts used to build the underlying SIDS and vaccination distributions, to randomly generate virtual population data, to fit the modified SCCS, to conduct all simulations, and to generate figures are all provided in the following github repository:

https://github.com/Jeffrey-S-Morris/SIDS-SCCS-Simulation